GENODERMATOSIS

Dr eman gomaa

Consultant dermatologist

BASIC CONCEPTS IN GENETICS	
Nuclear DNA	• Deoxyribonucleic acid (DNA) contained within nuclei of eukaryotic organisms • Packed into chromosomes • Selectively folded and unfolded to allow the expression of hundreds of genes at different times and in different cell types • Each somatic cell contains approximately 2 meters of DNA
Chromosome	• Organized structures of DNA and proteins that are found within the nucleus • Each diploid organism (including humans) has two sets of chromosomes, one from its father and one from its mother • Each pair of chromosomes is composed of two homologous chromosomes, one from each parent • Humans normally have 23 pairs of chromosomes: 22 pairs of autosomes and one pair of sex chromosomes (XX in females and XY in males) • Chromosomes X and Y share only two small regions, known as the pseudoautosomal regions
Karyotype	• Describes the chromosomal constitution of an individual (e.g. 46, XY or 46, XX)
Haploid cell	• Contains only one chromosome complement or "n" (for humans, n = 23) • Gametes (paternal and maternal) are haploid cells
Diploid cell	• Contains two chromosome complements or "2n" (for humans, 2n = 46) • Human cells (other than gametes) are diploid
Centromere	• A site of constriction in a chromosome that is essential for correct cell division • Depending on the position of the centromere, chromosomes can be: - *Metacentric*: centromere near the middle, two "arms" of similar length - *Submetacentric*: centromere between the center and end, two "arms" of somewhat different lengths - *Acrocentric*: centromere near the end, two "arms" of very different lengths • Composed of a series of histone proteins and repetitive DNA sequences (several thousand kilobases) • Divides the chromosome into two arms: the short (petite) arm or ***p***, and the long arm or ***q***
Telomere	• A region of repetitive DNA (TTAGGG sequence) and proteins at the ends of each chromosome • Plays a key role in the maintenance of chromosomal integrity and ensuring correct replication
Meiosis	• The process of cell division by which a diploid cell from the germline gives rise to haploid gametes via segregation and random assortment of homologous chromosomes • Results in combinations of chromosomes in the daughter cells that are different than that in the parental cell
Mitosis	• The process of somatic cell division of proliferating cells by which the genetic material of a cell is duplicated and equally distributed between the two resulting daughter cells • To achieve this duplication of genetic material, the chromosome replicates itself and forms two *sister chromatids*, which will each become the chromosome of a daughter cell
Locus	• The location of a particular sequence of DNA (e.g. a gene or DNA fragment) on a chromosome • Since we have two copies of each chromosome, we also have two copies of each locus, which can be identical or different
Gene	• A segment of DNA that encodes a protein (or occasionally an RNA chain) that serves a function
Allele	• Alternative forms of a particular gene or DNA sequence at a given locus • An individual has two alleles at each autosomal locus (one from each parent), and the locus/individual is referred to as: - *Homozygous* if the alleles are identical - *Heterozygous* if the alleles are different • Since males have only one copy of the X and Y chromosomes, they are *hemizygous* for loci on these two chromosomes
Polymorphic locus	• Exhibits at least two alleles with a frequency in the general population of >1% • Can be located within or outside of a gene
Single nucleotide polymorphism (SNP)	• A single nucleotide in a particular location within the genome that has at least two different forms with a frequency of >1% in the general population • Occurs on average every 1000 nucleotides
Mutation	• A change in the DNA sequence • Refers to variation resulting in a disease phenotype • Although any change in the DNA is technically a mutation, the term is often used to refer to a pathogenic variant that results in a disease phenotype
Genotype	• The alleles present at a specific locus (or in the whole genome) of an individual
Phenotype	• The manifestation of a particular genotype (in the context of the environment in which it is expressed) • Different genotypes are not always recognizable as distinct phenotypes • A change in a phenotype can be caused by variant alleles that are present in the general population (normal variation) or by pathogenetic variation that leads to a disease

PATTERNS OF INHERITANCE					
Pattern	Parents affected	Gender affected	Transmission	Recurrence risk	Risk factors
Autosomal recessive	No (carriers)	Both equally	Disease seen in siblings of proband, not in parents or offspring Usually only in one generation	1 in 4	Consanguinity, isolated population (e.g. geographically, linguistically)
Autosomal dominant	Yes*	Both equally	Disease seen in successive generations	1 in 2	*De novo* mutations
X-linked recessive	Mother a "carrier"*	Males have the "complete" disease Female "carriers" may have mild manifestations (e.g. in a mosaic pattern)**	No male-to-male transmission (but all daughters of an affected male are "carriers")	1 in 2 male children born to a female "carrier" will be affected (and 1 in 2 of her female children will be carriers)	*De novo* mutations
X-linked dominant	Yes*	Predominantly females if lethal in males during embryonic development; otherwise milder in females (often with a mosaic pattern of skin lesions) and more severe in males	Affected males have: (1) no affected sons; and (2) all daughters affected No male-to-male transmission	1 in 2 children born to affected female; may spontaneously abort male fetuses if "male-lethal" condition	*De novo* mutations

ichthyosis

- Group of disorder of keratinization →
- Ch.by abnormal differentiation , desquamation .→generalized scaling ,xerosis of skin

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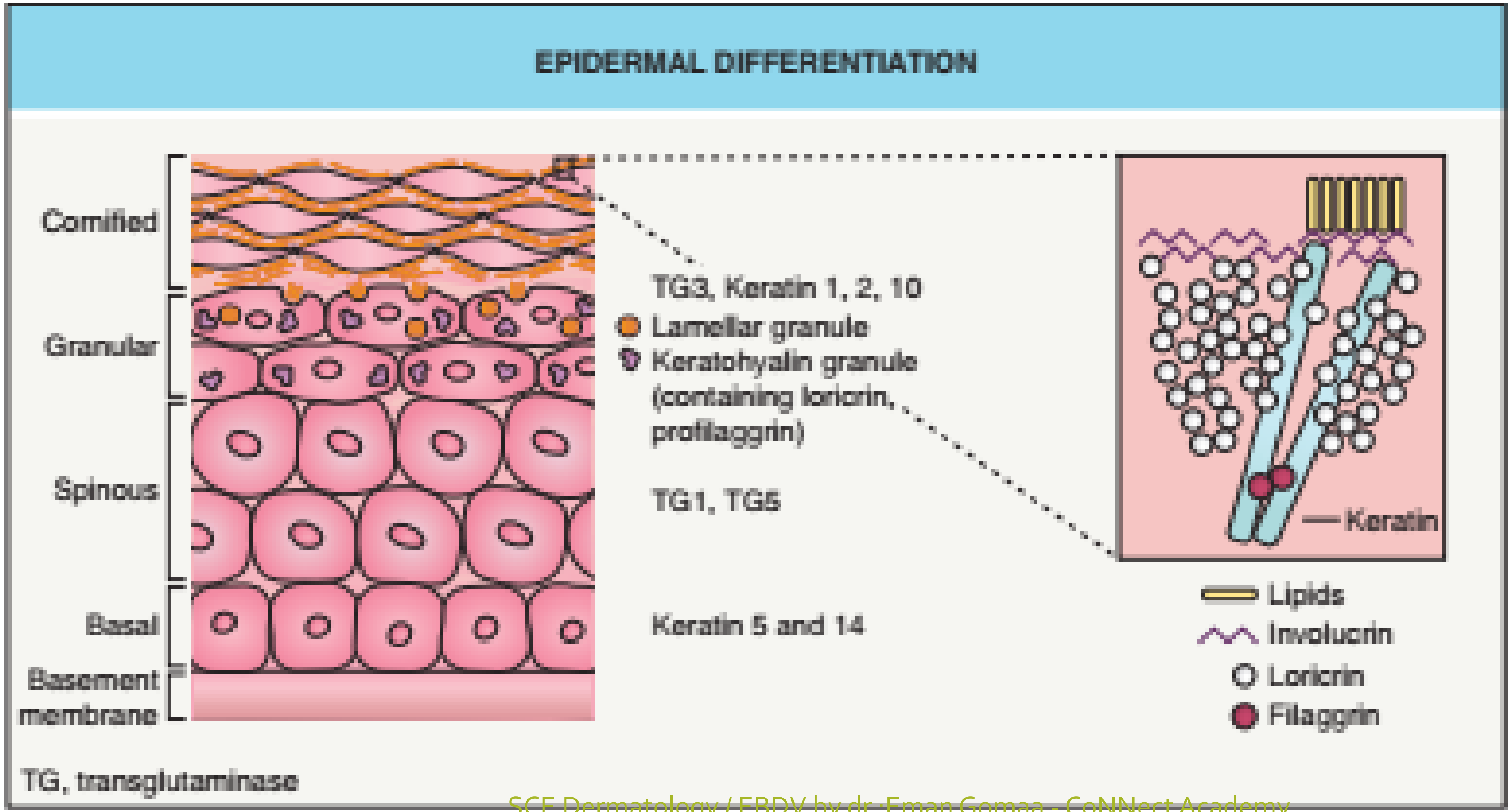
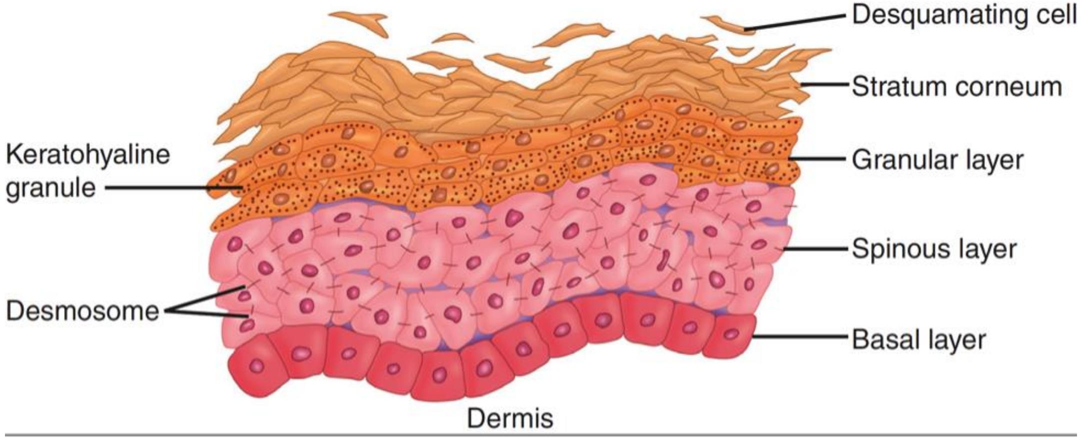
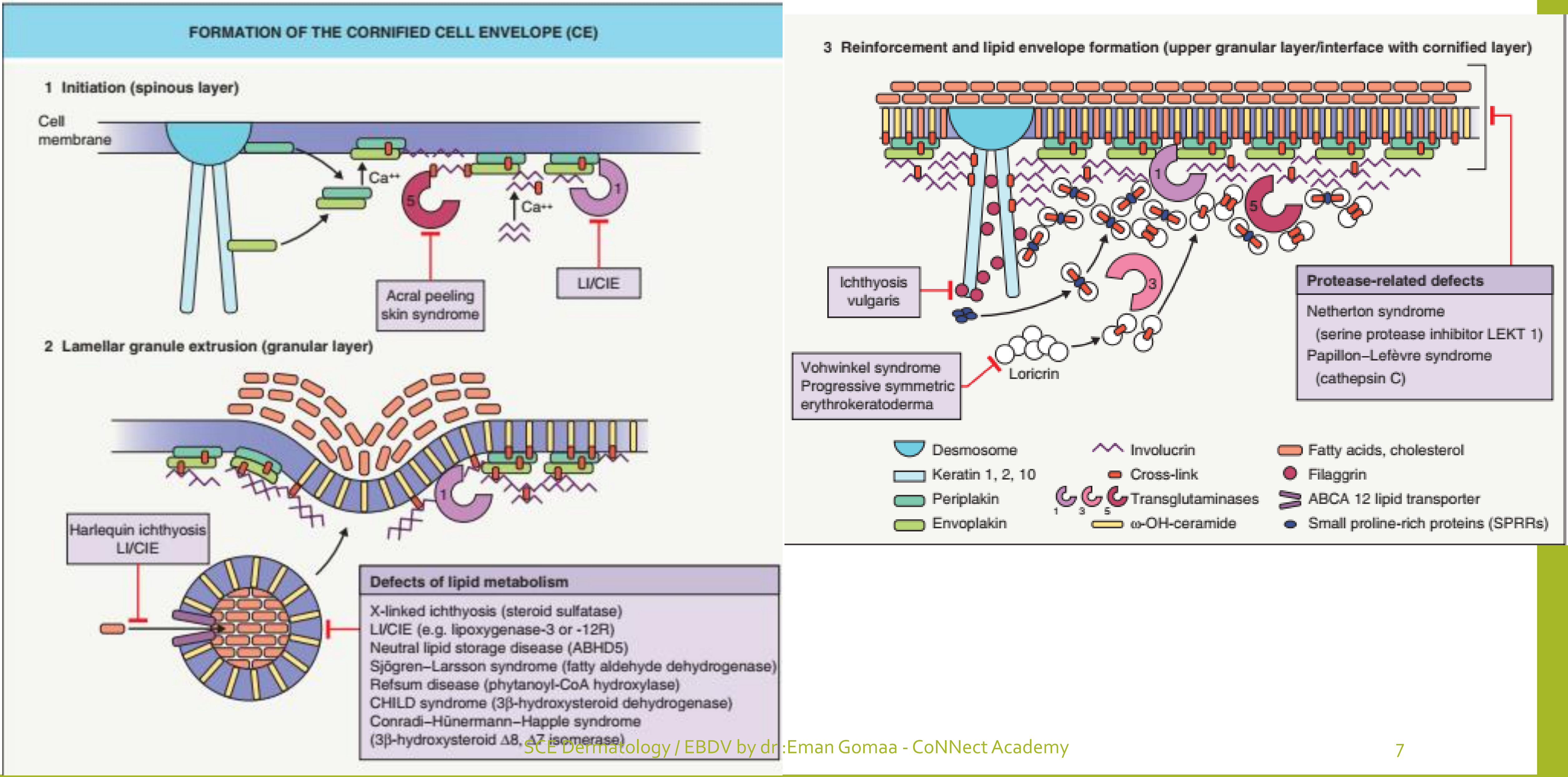


EPIDERMAL DIFFERENTIATION
Cornified
Granular
Spinous
Basal
Basement membrane
TG3, Keratin 1, 2, 10
Lamellar granule
Keratohyalin granule (containing loricrin, profilaggrin)
TG1, TG5
Keratin 5 and 14
Keratin
Lipids
Involucrin
Loricrin
Filaggrin
TG, transglutaminase

Stratum granulosum

- Contain basophilic keratohyaline granules responsible for protein compartment.
- Lamellar body(Odland body) for lipid compartment .

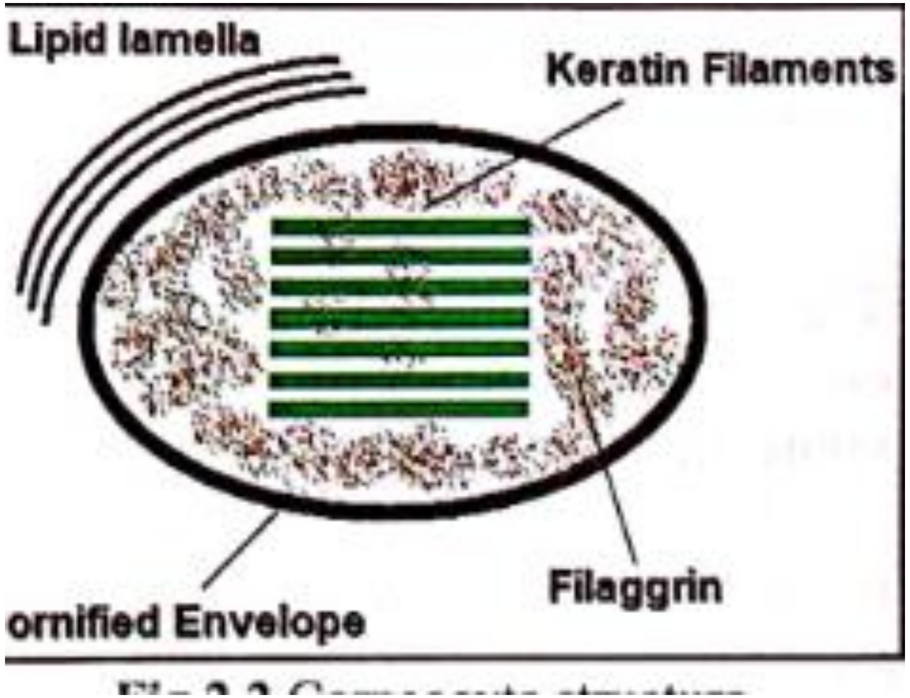




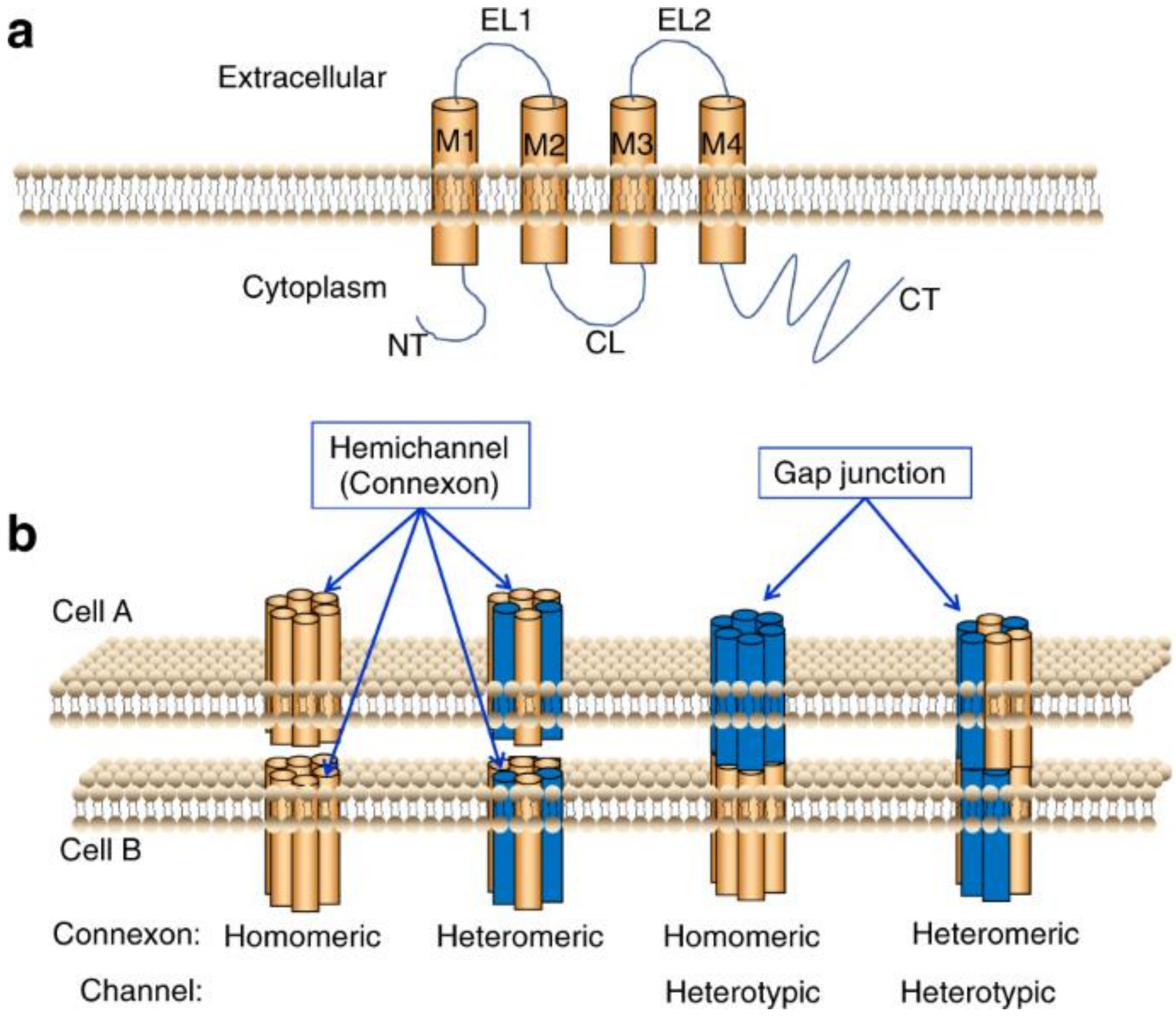
FORMATION OF THE CORNIFIED CELL ENVELOPE (CE)
1 Initiation (spinous layer)
Cell membrane
Ca++
Ca++
5
1
Acral peeling skin syndrome
LI/CIE
2 Lamellar granule extrusion (granular layer)
1
Harlequin ichthyosis LI/CIE
Defects of lipid metabolism
X-linked ichthyosis (steroid sulfatase)
LI/CIE (e.g. lipoxygenase-3 or -12R)
Neutral lipid storage disease (ABHD5)
Sjögren–Larsson syndrome (fatty aldehyde dehydrogenase)
Refsum disease (phytanoyl-CoA hydroxylase)
CHILD syndrome (3β-hydroxysteroid dehydrogenase)
Conradi–Hünermann–Happle syndrome
(3β-hydroxysteroid Δ8, Δ7 isomerase)
3 Reinforcement and lipid envelope formation (upper granular layer/interface with cornified layer)
1
5
3
Ichthyosis vulgaris
Vohwinkel syndrome
Progressive symmetric erythrokeratoderma
Loricrin
Protease-related defects
Netherton syndrome
(serine protease inhibitor LEKT 1)
Papillon–Lefèvre syndrome
(cathepsin C)
Desmosome
Keratin 1, 2, 10
Periplakin
Envoplakin
Involucrin
Cross-link
Transglutaminases
1 3 5
ω-OH-ceramide
Fatty acids, cholesterol
Filaggrin
ABCA 12 lipid transporter
Small proline-rich proteins (SPRRs)

- Plasma membrane replased by cornified envelop
- Cytoplasm is filled with keratin
- The structure and organization of SC is described as

(**brick and morter)**



Gap junction tranmembran channel



Development of startum corneum

- Granular layer has 2 types of granules

Keratohyaline Granule (Protein)	Lamellar body (Lipid)
Filaggrin Transgluminas 1 Cross linking protein (cornified envelop) Involucrin Loricrin	ABCA12 Extracellularly transport lipid

Extracellular lipid

1. Ceramid 50%
2. Cholesterol 35-40%
3. FFA 15%

Lipid prevent excessive loss of water

Degradation of SC

Filaggrin → natural moisturizing factor
cholesterol sulfate → inhibit activity of protease → at lower epidermis .
Cholesterol sulafate decrease at upper dermis → increase protease activity →degradation of desmosome

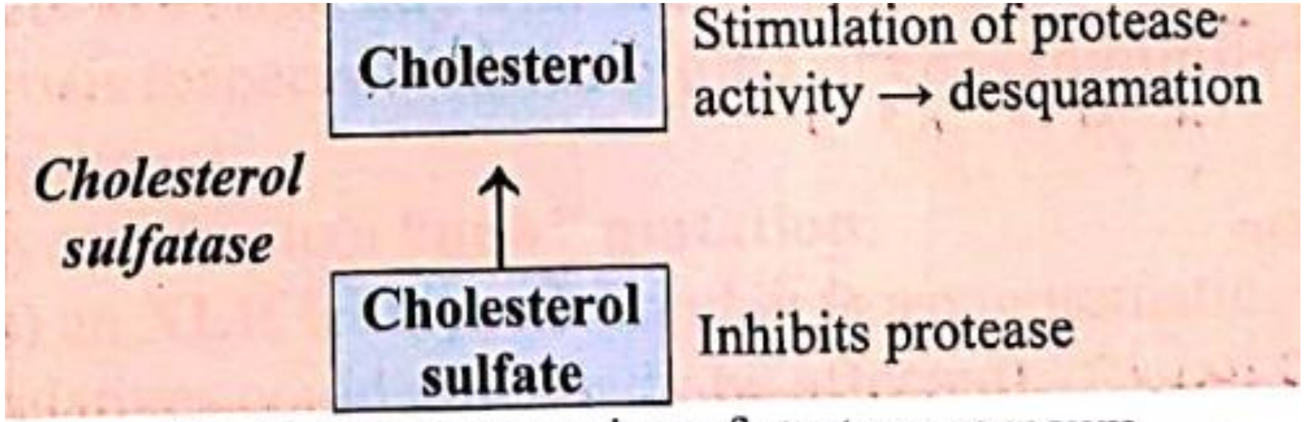


Fig 2.4 Desquamation of stratum corneum.

Componant involved in ichthyosis

1. KIF (K1- K10
2. KERATOYALINE GRANULES
3. FILLAGRIN
4. NMF
5. TG1
6. CE
7. LAMELLAR BODY
8. LIPID PERCURSOR (CHOLESTEROL SULFATE
9. STEROID SULFATASE
10. ABCA12 / CA
11. SERINE PORETEASE

- ichthyosis
  - congenital
  - acquired

Acquired ichthyosis:

1. Malignancy
2. Malabsorbtion
3. Inflammatory :sarcoidosis
4. Infection:leprosy , AIDs
5. Chronic disease:hepatic , renal , thyroid
6. Drugs :hypocholesterolemic agent nicotinic acid

Congenital Itchthyosis

Common	AR congenital	keratinopathic	synthetic
Ichthyosis vulgraris X-linked recessive ICH (steroid sulfatase def.)	CIE Lamellar ic Harlequin	Epidermolytic Ich Ich –hystrex Ich-confetdi Superficial epidermolytic Ich	Nethroten sjogren larsin Refsum trichothiodestrophy

Important points in ichthyosis

- Mode of inheritance
- Defect
- Scale type
- Scale distribution

Investigation
Hp
Microscopy
Laboratory study
immunohisteochemistr
y

Treatment
Reduction of scaling
1. Topical :emollient ,keratolytic
2. Systemic :
Retinoids

FEATURES OF SELECTED ICHTHYOSES AND ERYTHROKERATODERMAS							
Diagnosis	Gene	Mode of inheritance	Incidence	Onset	Primary cutaneous features	Associated features	Diagnosis (in addition to molecular testing*)
Common ichthyoses							
Ichthyosis vulgaris	*FLG*	Autosomal semi-dominant	1:100–1:250	Infancy/ childhood	Fine, adherent scales on extremities and trunk with sparing of flexures; larger scales on lower legs; hyperlinear palms/soles, furrowed heels	Keratosis pilaris; atopic dermatitis	Clinical; diminished/ absent stratum granulosum; absent/ reduced filaggrin immunostaining
Steroid sulfatase deficiency (X-linked recessive ichthyosis)	*STS* (deleted in ~90% of patients)	X-linked recessive	1:2000–1:9500 boys/men	Infancy	Fine to large, dark, adherent scales on extremities, trunk, neck, and lateral face; sparing of skin folds	Corneal; opacities; cryptorchidism *Female carriers*: corneal opacities; prolonged labor with affected child	FISH or targeted/ whole genome microarray to identify the gene deletion; increased plasma (hydroxy) cholesterol sulfate; lipoprotein electrophoresis (increased mobility of β-fraction); decreased steroid sulfatase activity in leukocytes

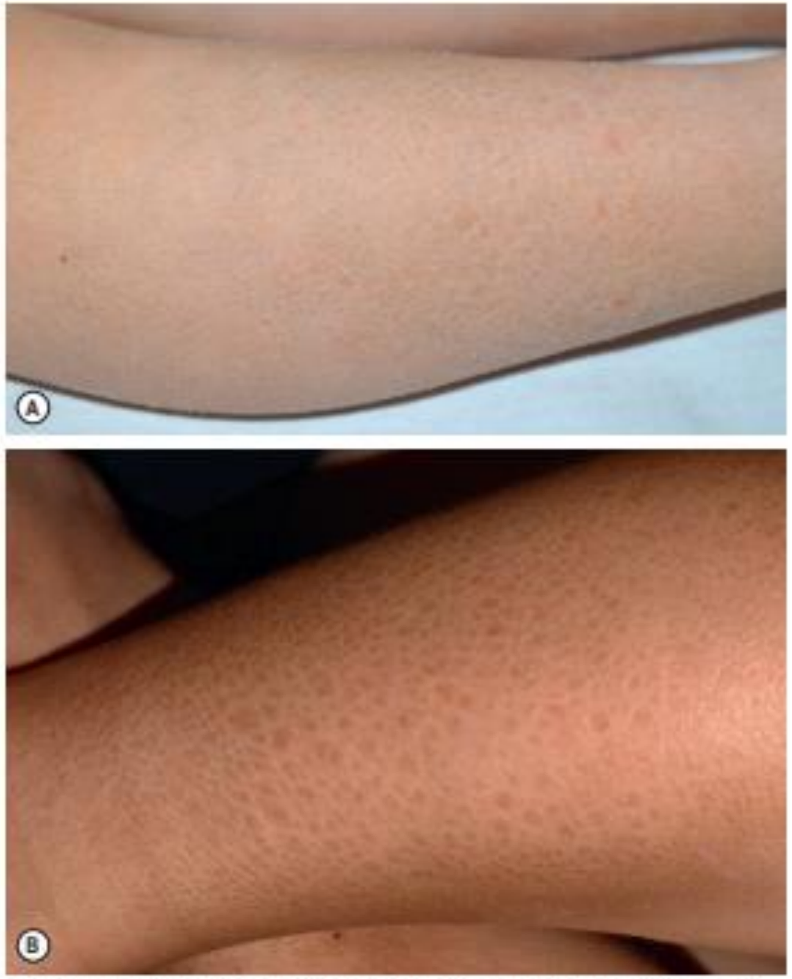


Fig. 57.2 Ichthyosis vulgaris. Fine, whitish, flaky scales (**A**) and larger, darker scales (**B**) on the legs of children with skin phototypes II and IV, respectively. *A, Courtesy, Angela Hernández-Martín, MD; B, Courtesy, Julie V Schaffer, MD.*

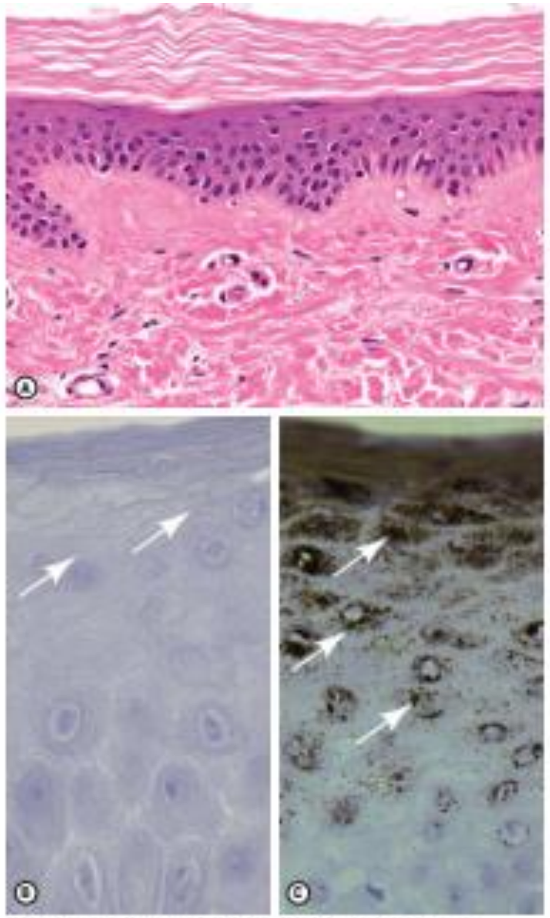


Fig. 57.4 Histologic features of ichthyosis vulgaris. A Orthokeratotic hyperkeratosis and a diminished granular layer (arrows) are evident. Immunostaining with an anti-filaggrin antibody demonstrates a complete absence of filaggrin in an ichthyosis vulgaris patient homozygous for a deleterious *FLG* mutation (**B**), while filaggrin is strongly expressed throughout the granular layer (arrows) in normal skin (**C**).

Fig. 57.3 Hyperlinear palms in ichthyosis vulgaris.

Icthyosis vulgaris
Ich – simplex/ AD

Most common 4:1000

Infancy & childhood equal sex

Mode :- AD

Defect:- Filaggrin

Fine white flaky scales → extensors sparing flexors.

Symptoms:- pruritis improving in summer & worse in cold

- Association:-
  - Atopci dermatitis
  - Palmoplantar hyperliniarity
  - Keratosis pilaris
- Couse:- improves with age.
- H.P:- hyperkeratosis & hypogranulosis.
- E.M:- no keratohyaline granules.
- Immunohistochemistry → decreased flaggrin stain

- TTT:-

  topical emollient

  Keratolytic.

  Retinoids.

Fig. 57.5 Steroid sulfatase deficiency. Large light-brown (A) and more prominent dark-brown (B) scales on the lower legs. C Smaller dark-brown scales on the trunk, with sparing of the skin folds. D Dark scales on the neck, sometimes referred to as a "dirty neck". A–C, Courtesy, [illegible]

Nonsyndromic autosomal recessive congenital ichthyoses							
Lamellar ichthyosis (LI), congenital ichthyosiform erythroderma (CIE)	**LI > intermediate/CIE:** *TGM1* *ABCA12* *CYP4F22* *SDR9C7* *SULT2B1*** **CIE > intermediate/LI:** *ALOX12B* *ALOXE3* *NIPAL4/ICHTHYIN* *PNPLA1* *CERS3* **Childhood-onset intermediate:** *LIPN*	Autosomal recessive†	LI: 1 : 200 000–1 : 300 000 CIE: 1 : 100 000–1 : 200 000	Birth	Frequently collodion membrane at birth, then a generalized distribution; variable palm/sole involvement LI: large, thick, plate-like brown scales; absent or mild erythroderma CIE: fine, white scales; erythroderma	Heat intolerance; frequent (LI) or variable (CIE) scarring alopecia and ectropion > eclabium	Clinical; transglutaminase-1 immunostaining or *in situ* assay in skin biopsy specimen
Harlequin ichthyosis	*ABCA12* *(KDSR)*	Autosomal recessive	Rare	Birth	Very thick, yellow-brown plates of scale that tightly encase the neonate; large, deep, bright red fissures; survivors develop severe CIE-like phenotype	Premature delivery; often neonatal death due to sepsis or respiratory insufficiency; extreme ectropion, eclabium, and ear deformities	Clinical

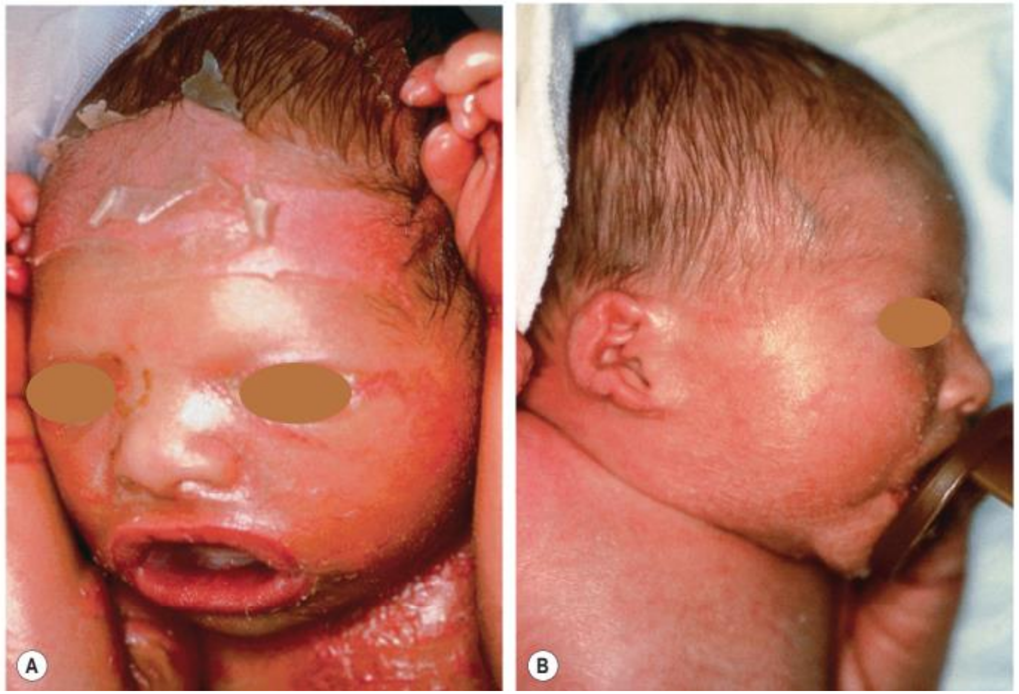


Fig. 57.10 Collodion baby. A Day 1 with eclabium. **B** Day 8 with erythema and diffuse mild scaling and misshapen ears.

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Fig. 57.11 Lamellar ichthyosis. **A** Large, plate-like scales on the lower extremities forming a mosaic pattern. **B** Brown scales of various sizes with accentuation on the neck and in the axilla. **C** Obvious ectropion as well as plate-like scales. *B, Courtesy, Julie V Schaffer, MD.*

ig. 57.12 Congenital ichthyosiform erythroderma. A Intense redness and ne, flaky, white scale on the trunk and arms. Close-up of fine white (B) and oarser yellowish (C) scale in a background of prominent erythema. *A, Courtesy, S.*

Fig. 57.13 Harlequin ichthyosis. Severe hyperkeratosis with fissuring as well as eclabium and ectropion.
Reproduced from Morillo M, Novo R, Torrelo A, et al. Feto arlequin. Actas Dermosifiliogr. 1999;90:185–7.

FEATURES OF SELECTED ICHTHYOSES AND ERYTHROKERATODERMAS							
Diagnosis	Gene	Mode of inheritance	Incidence	Onset	Primary cutaneous features	Associated features	Diagnosis (in addition to molecular testing*)
Keratinopathic ichthyoses (also includes ichthyosis with confetti – see text)							
Epidermolytic ichthyosis	*KRT1* *KRT10*	Autosomal dominant (very rarely autosomal recessive)	1:200000–1:350000	Birth	*At birth:* erythroderma, blistering, erosions *Later:* hyperkeratosis with cobblestone pattern (most prominent over joints), ridging of the flexures; generalized or localized; variable degree of palmoplantar involvement and blistering	Frequent skin infections; malodor; gait and posture abnormalities	Clinical; histopathology
Superficial epidermolytic ichthyosis	*KRT2*	Autosomal dominant	Rare	Birth	Erythroderma and blistering at birth; later, hyperkeratosis with flexural accentuation, "molting" of the skin; palms and soles spared		Clinical; histopathology
Ichthyosis hystrix Curth–Macklin	*KRT1*	Autosomal dominant	Rare	Birth	Mild to severe, mutilating palmoplantar keratoderma; hyperkeratosis with verrucous, cobblestone, or hystrix-like pattern on extremities and trunk	Pseudoainhum; digital contractures	Clinical; electron microscopy

Fig. 57.6 Epidermolytic ichthyosis. A Erythroderma with fine scaling and erosions on the face and thigh in an infant. B Later development of marked hyperkeratosis with focal erosions. C, D Corrugated hyperkeratosis in the antecubital fossa. E Hyperkeratosis with a cobblestone pattern on the dorsal hand. F Severe palmoplantar keratoderma with digital contractures in a patient with a KRT1 mutation.

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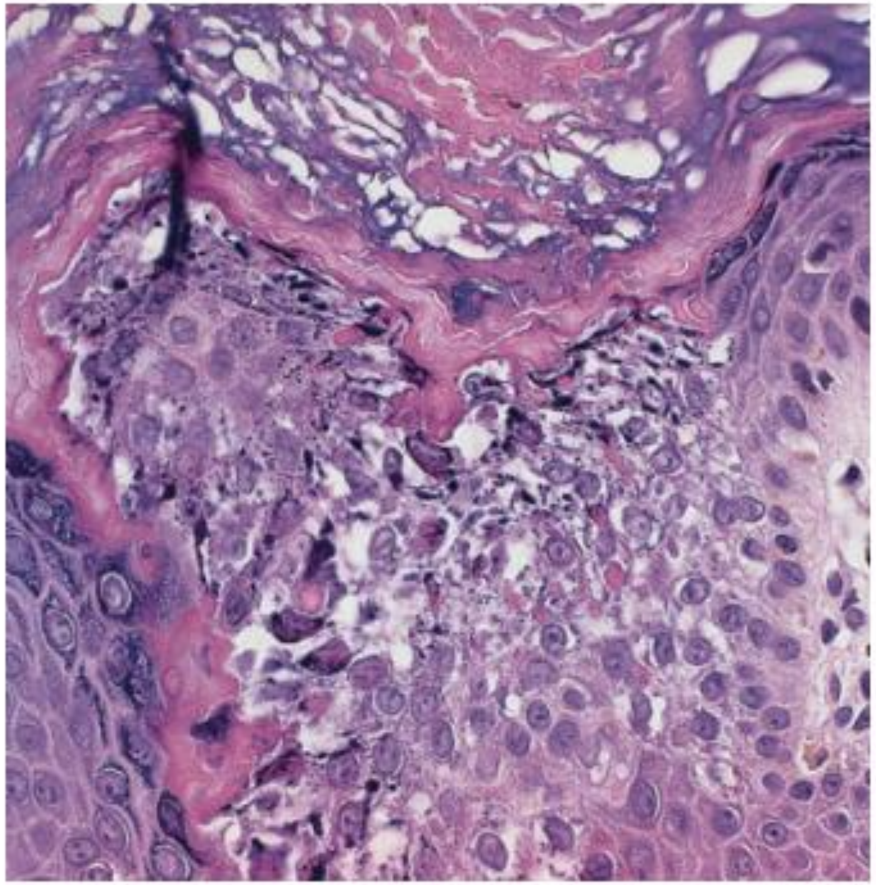


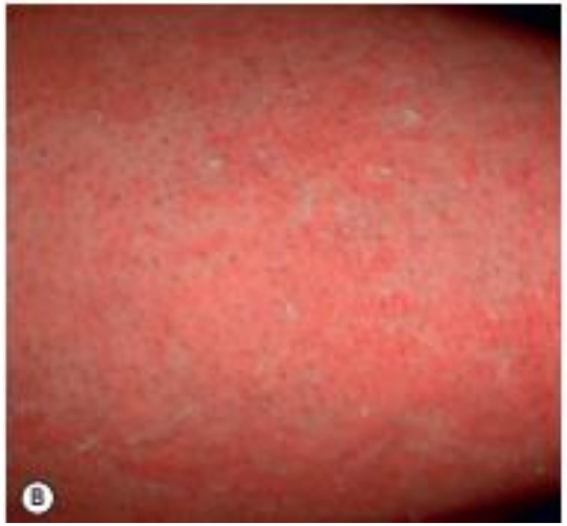
Fig. 57.7 Histology of epidermolytic ichthyosis. Striking ortho-hyperkeratosis, intracellular vacuolization of keratinocytes, and a prominent granular layer with clumped keratin filaments are

ICH-HYSTRIX CURTH MACLIN

Fig. 57.9 Ichthyosis with confetti. Numerous small, confetti-like "islands" in the background of ichthyosiform erythroderma reflect revertant mosaicism. *Courtesy, Antonio Torrelo, MD.*

Syndromic ichthyosis

Syndromic ichthyoses							
Netherton syndrome	*SPINK5*	Autosomal recessive	1 : 50 000	Birth/ infancy	Congenital erythroderma and peeling; two principal phenotypes: ichthyosis linearis circumflexa and CIE-like; pruritus and eczematous plaques	Trichorrhexis invaginata and other hair shaft abnormalities; highly elevated serum IgE; neonatal temperature instability, electrolyte imbalance, failure to thrive; recurrent infections; food and other allergies; nonspecific aminoaciduria	Clinical; psoriasiform histopathology; trichoscopy, light microscopic hair shaft analysis; serum IgE measurement
Sjögren–Larsson syndrome	*ALDH3A2*	Autosomal recessive	<1 : 100 000	Birth	Fine to plate-like scaling or non-scaling hyperkeratosis; favors abdomen, neck, flexures; lichenification; pruritus	Spastic di- and tetraplegia; perifoveal glistening white dots; intellectual disability; white matter disease of the brain	Fatty aldehyde hydrogenase activity assay in cultured fibroblasts (limited availability)



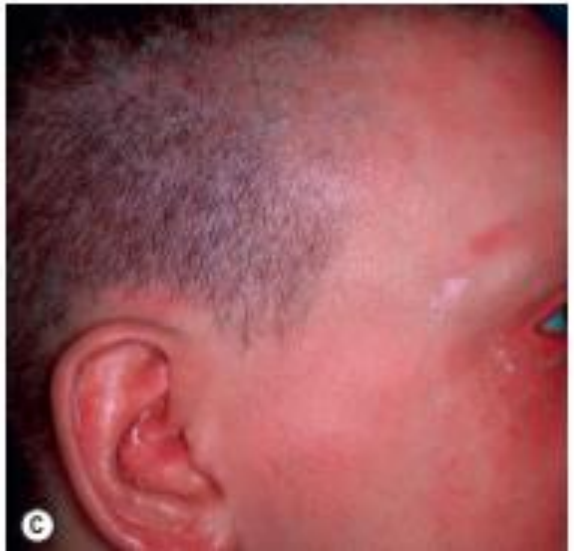


Fig. 57.14 Netherton syndrome. A Generalized involvement with features of congenital ichthyosiform erythroderma. **B** Close-up showing "peeling" quality of the scale. **C** Short, thin hair on the scalp, sparse eyebrows and a lack of

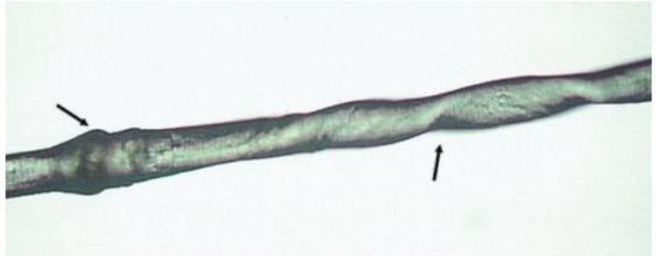


Fig. 57.16 Abnormal hair shaft in Netherton syndrome. Trichorrhexis invaginata with its ball-and-socket appearance (left arrow) and twist in the hair shaft (right arrow).

Fig. 57.15 Ichthyosis linearis circumflexa. Note the double-edged scale.

Fig. 57.17 Sjögren–Larsson syndrome. Yellowish-brown hyperkeratosis with accentuated skin markings. *Courtesy, Julie V Schaffer, MD.*

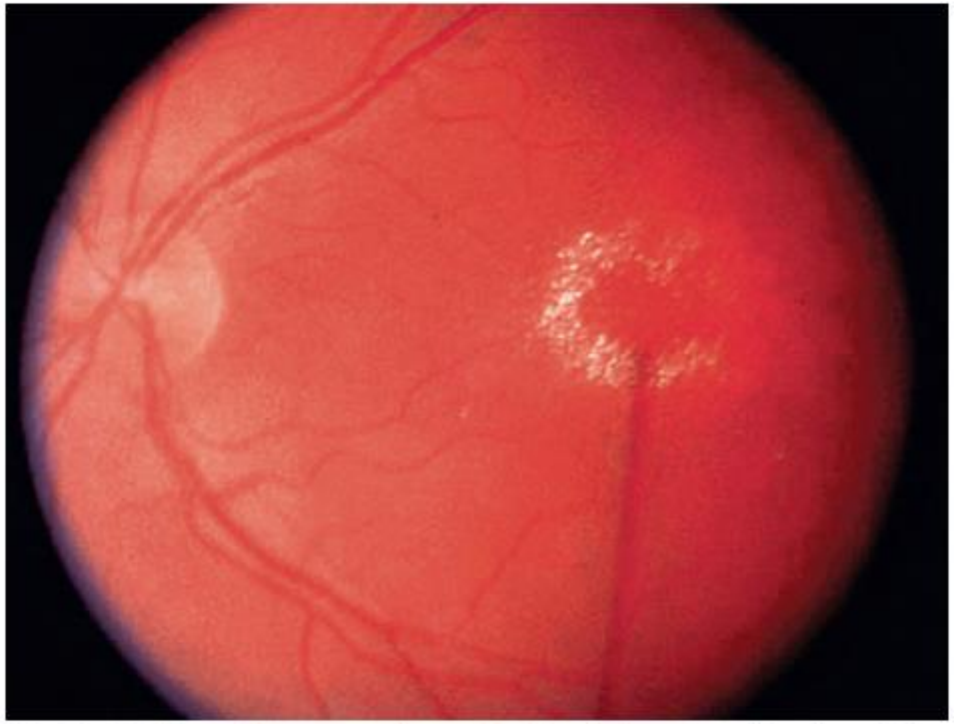


Fig. 57.18 Sjögren–Larsson syndrome. Perifoveal glistening white dots of the ocular fundus. *Courtesy, WB Rizzo, MD.*

FEATURES OF SELECTED ICHTHYOSES AND ERYTHROKERATODERMAS							
Diagnosis	Gene	Mode of inheritance	Incidence	Onset	Primary cutaneous features	Associated features	Diagnosis (in addition to molecular testing*)
Neutral lipid storage disease with ichthyosis	*ABHD5 (CGI-58)*	Autosomal recessive	Rare	Birth	Generalized fine, white scales with variable erythema	Developmental delay; hepatomegaly; myopathy; hearing impairment; cataracts; lipid-containing vacuoles in leukocytes	Peripheral blood smear to detect leukocyte lipid vacuoles; oil stains of frozen skin biopsy specimens
Trichothiodystrophy with ichthyosis	*ERCC2* *ERCC3* *GTF2H5* Often not associated with ichthyosis or photosensitivity: *MPLKIP/ C7ORF11/TTDN1* *RNF113A*	Autosomal recessive X-linked	Rare	Birth	Generalized scaling with absent or minimal erythroderma, except during infancy; collodion membrane in ~30% of patients	Brittle hair and nails; photosensitivity in ~50% of patients; intellectual disability; gonadal abnormalities; short stature	Clinical; tiger tail pattern by microscopic hair shaft examination under polarizing light; analysis of hair sulfur/cysteine content

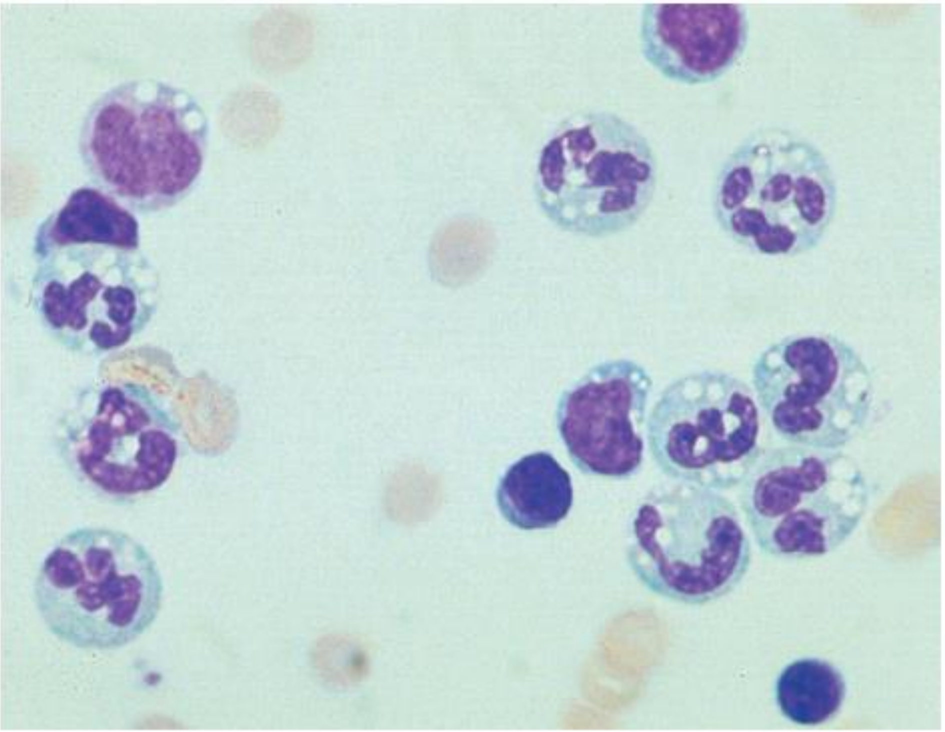
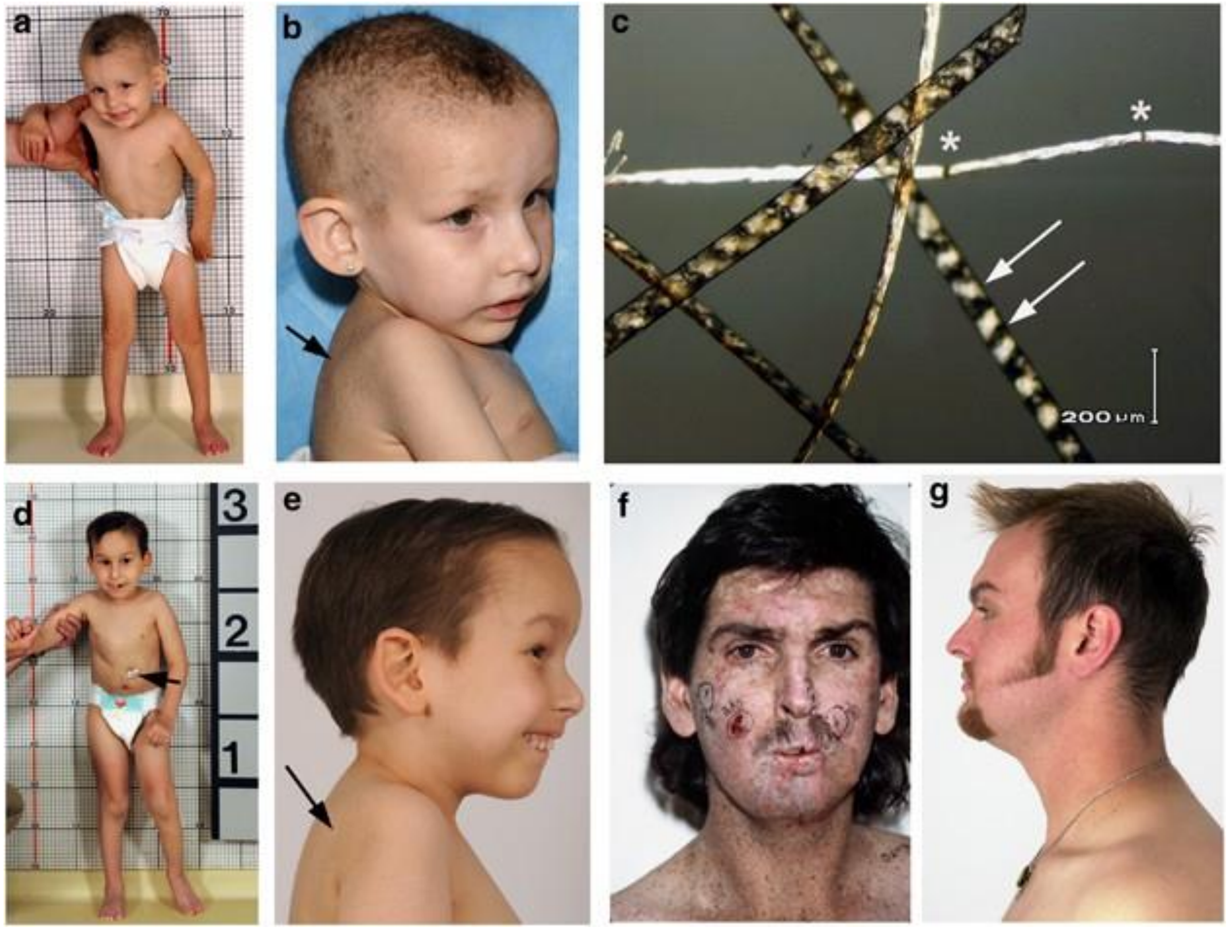


Fig. 57.19 **Neutral lipid storage disease.** Lipid droplets in circulating granulocytes and monocytes but not lymphocytes or erythrocytes. *Courtesy, Mary*



trichthiodystrophy

- 5- Child syndrome :
  - Conginital hemidysplasia
  - Ichthyosiform erythroderma
  - Limp defect
  - Systemic feature

Syndromic X-linked dominant ichthyosiform conditions							
CHILD syndrome	*NSDHL*	X-linked dominant	Rare	Birth	At birth, unilateral erythema and waxy, yellowish adherent scale; later, verrucous; hyperkeratosis of variable extent, with affinity for skin folds	Ipsilateral skeletal hemidysplasia, organ hypoplasia	Clinical

- 6-conradi happle man syndrome

Conradi–Hünermann–Happle syndrome	*EBP*	X-linked dominant	Rare	Birth	At birth, ichthyosiform erythroderma with feathery, adherent scale along lines of Blaschko; later, follicular atrophoderma along lines of Blaschko	Unilateral cataracts; chondrodysplasia punctata (during infancy); asymmetric skeletal abnormalities; patchy scarring alopecia	Clinical; epiphyseal stippling on X-rays during infancy; accumulation of plasma 8(9) cholesterol

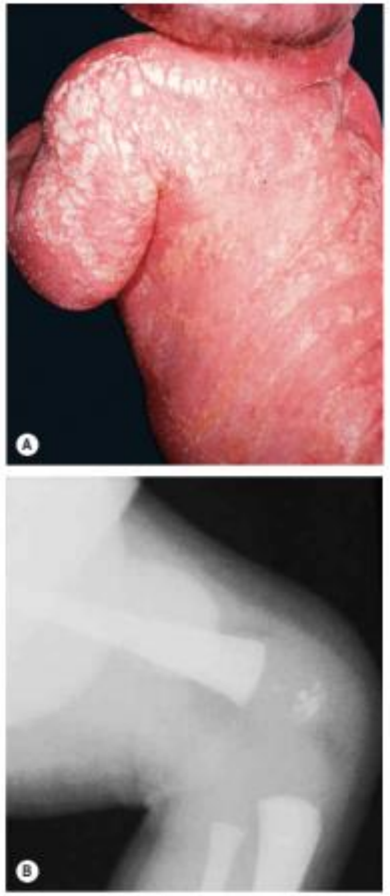


Fig. 57.24 Conradi–Hünermann–Happle syndrome. A Erythroderma and linear streaks and whorls of hyperkeratosis. **B** Epiphyseal stippling at the knee. *A, Courtesy, R Happle, MD; B, Courtesy, Jean L Bolognia, MD.*

NEONATAL PRESENTATION OF SELECTED ICHTHYOSES AND RELATED DISORDERS					
Disorder	Collodion membrane	Bullae/blisters/erosions	Erythroderma	Generalized ichthyosis	Peeling/exfoliation
Epidermolytic ichthyosis		+	+	(+)	(+)
Superficial epidermolytic ichthyosis		(+)	(+)	(+)	+
Lamellar ichthyosis	+		(+)	+	
Congenital ichthyosiform erythroderma	+		+	+	
Self-improving collodion ichthyosis	+				
Sjögren–Larsson syndrome	(+)		+	+	
Neutral lipid storage disease	(+)		+	+	
Trichothiodystrophy	(+)		+	+	
Netherton syndrome			+	+	+
Steroid sulfatase deficiency			+	(+)	+
Peeling skin syndromes (see Table 57.5)			(+)	(+)	+
Conradi–Hünermann–Happle syndrome	(+)		+ (lines of Blaschko)	+ (lines of Blaschko)	
CHILD syndrome			Unilateral	Unilateral*	
KID syndrome			+	(+)	

*Some attenuated or band-like involvement may occasionally be appreciated on the contralateral side.

Erythrokeratoderma

Erythrokeratodermas							
Erythrokeratodermia variabilis	*GJB3* *GJB4* *GJA1* (less common)	Autosomal dominant (very rarely autosomal recessive)	Rare	Birth/ infancy	Transient, variable erythematous patches; more stable, geographic, hyperkeratotic plaques over knees, elbows, Achilles tendons, extremities, buttocks, lateral trunk; face and scalp often spared; less common generalized hyperkeratosis; palmoplantar keratoderma in ~50% of patients	Burning or stinging sensation preceding or accompanying erythema	Clinical, particularly history or presence of transient erythema
Progressive symmetric erythrokeratoderma	No consistently implicated genes (*LOR*) (*GJB4*) (*KRT83*) (*KDSR* – see Table 57.5)	Autosomal dominant Autosomal recessive	Rare	Infancy/ childhood	Fixed, slowly progressive, erythematous, hyperkeratotic plaques with sharp, figurate borders; on cheeks, over knees, elbows, extremities, rarely trunk; palmoplantar keratoderma common		Clinical

Fig. 57.20 Erythrokeratodermia variabilis. A Transient erythematous patches and generalized brownish hyperkeratosis. **B** Symmetrically distributed hyperkeratotic plaques with hypertrichosis. **C** Brown hyperkeratotic plaques with geographic borders in a 7-year-old child. C Courtesy, Julie V Schaffer MD

Fig. 57.21 Progressive symmetric erythrokeratoderma. Well-demarcated, symmetrically distributed hyperkeratotic plaques on the buttocks and lower extremities; note the white scale with underlying erythema and hyperpigmentation. *Courtesy, Antonio Torrelo, MD.*

KID SYNDROME

Diagnosis	Gene	Mode of inheritance	Incidence	Onset	Primary cutaneous features	Associated features	Diagnosis (in addition to molecular testing*)
Keratitis–ichthyosis–deafness syndrome	*GJB2* (*GJB6* – single reported patient)	Autosomal dominant	Rare	Birth/ infancy	Transient neonatal erythroderma; erythematous, hyperkeratotic plaques with well-demarcated, borders on face and extremities; follicular keratoses; thickening of the skin with an appearance of "coarse-grained leather"; stippled palmoplantar keratoderma	Congenital sensorineural hearing impairment; progressive keratitis with corneal neovascularization, conjunctivitis; recurrent mucocutaneous infections, especially with *Candida albicans*; increased susceptibility to oral and cutaneous squamous cell carcinoma; nail, hair and dental anomalies; very rare fatal form with respiratory problems	Clinical

Fig. 57.22 KID syndrome. A Diffusely thickened, hyperkeratotic skin with a coarse-grained texture and an erythematous component; note the characteristic radial furrows around the mouth. **B** Bright pink, keratotic periorificial plaques with crusting. Palmar **(C)** and plantar **(D)** keratoderma with a grainy surface. *A, C, Courtesy, Julie V Schaffer, MD; B, Courtesy, J Tercedor, MD; D, Courtesy, Richard Antaya, MD.*

APPROACH TO THE NEONATE OR INFANT WITH SIGNS OF ICHTHYOSIS AND DIAGNOSTIC CLUES

History
- Prenatal/birth history: prenatal diagnoses (e.g. heart defects), prolonged labor (suggestive of steroid sulfatase deficiency), prematurity (characteristic of harlequin ichthyosis, ichthyosis with prematurity)
- Consanguinity makes an autosomal recessive ichthyosis more likely
- Family history of epidermal nevi, ichthyosis, or erythrokeratoderma

Physical examination
- Types of skin findings (see Table 57.2): collodion membrane, bullae/erosions, erythroderma, peeling, diffuse scaling, circumscribed areas of erythema/hyperkeratosis
- Distribution of skin findings: generalized, unilateral, following lines of Blaschko, regional
- Evaluate the scalp hair, eyebrows, and eyelashes (including with trichoscopy); nails; and mucous membranes
- Evaluate for microcephaly, dysmorphic facies, asymmetry, contractures, hepatosplenomegaly, and other extracutaneous abnormalities

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Basic laboratory evaluation and additional assessment
- Complete blood count, electrolytes, hepatic panel, immunoglobulin levels (including IgE)
- Peripheral blood smear to evaluate for vacuoles in leukocytes
- Hearing screen (e.g. via the auditory brainstem response or oto-acoustic emission test)
- Ophthalmologic examination
- Consider X-rays to evaluate epiphyses, especially if ichthyosiform lesions are unilateral or in a mosaic distribution pattern
- Consider a skin biopsy, in particular if bullae/erosions (to assess for epidermolysis), if suspect ichthyosis vulgaris (to assess for a diminished granular layer; more evident in severe forms), or if genetic testing is not possible

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Consider genetic testing
- A multigene panel is preferable over single gene testing if the diagnosis is not evident based on clinical and laboratory findings
- FISH and microarray are alternatives if suspect steroid sulfatase deficiency

Other possible diagnostic testing (see Tables 57.1, 57.4, & 57.5)
- Following resolution of the collodion membrane if present, immunohistochemical staining of a skin biopsy specimen for transglutaminase-1, LEKTI, or filaggrin
- Lipid antibodies (e.g. adipophilin) or stains (e.g. oil red O, Sudan III) to detect cytoplasmic lipid droplets in paraffin-embedded or fresh frozen skin sections, respectively, if suspect neutral lipid storage disease
- Fibroblast or keratinocyte cultures of skin biopsy specimen to measure FALDH activity or check for elevated free fatty alcohols if suspect Sjögren–Larsson syndrome
- Electron microscopy of a skin biopsy specimen using ruthenium tetroxide fixation, to assess for abnormalities in the epidermal lipid system